Event Hub(Apache Kafka)

Saturday, 21 February 2026 10:57 PM

Azure Event Hubs: A Managed Event Streaming Service

Introduction

Azure Event Hubs is a fully managed, real-time data ingestion service provided by Microsoft Azure. It is designed to handle massive streams of events from diverse sources, acting as a highly scalable event streaming platform. Event Hubs operates on the fundamental principle of **publish-subscribe (pub-sub)**, also known as the **producer-consumer model**.

The Producer-Consumer Model

In the context of Event Hubs, there are two primary roles:

- **Producer**: An entity that sends events to an Event Hub. Producers can be any application, device, or service that generates data.
- **Consumer**: An entity that reads and processes events from an Event Hub. Consumers can be various analytics services, storage solutions, or other applications.

Example: Taxi Booking System

Consider an organization operating a web application for a taxi booking system, similar to Uber. In this scenario:

1. The **web application** acts as a **producer**. When a user interacts with the application, for instance, by hitting the "book" button, an **event** is created.
2. This event, containing information about the booking, is then sent to Azure Event Hubs. Event Hubs acts like a channel, storing these events in an ordered sequence.
3. For every booking action, a new event is generated and stored in chronological order within the Event Hub.

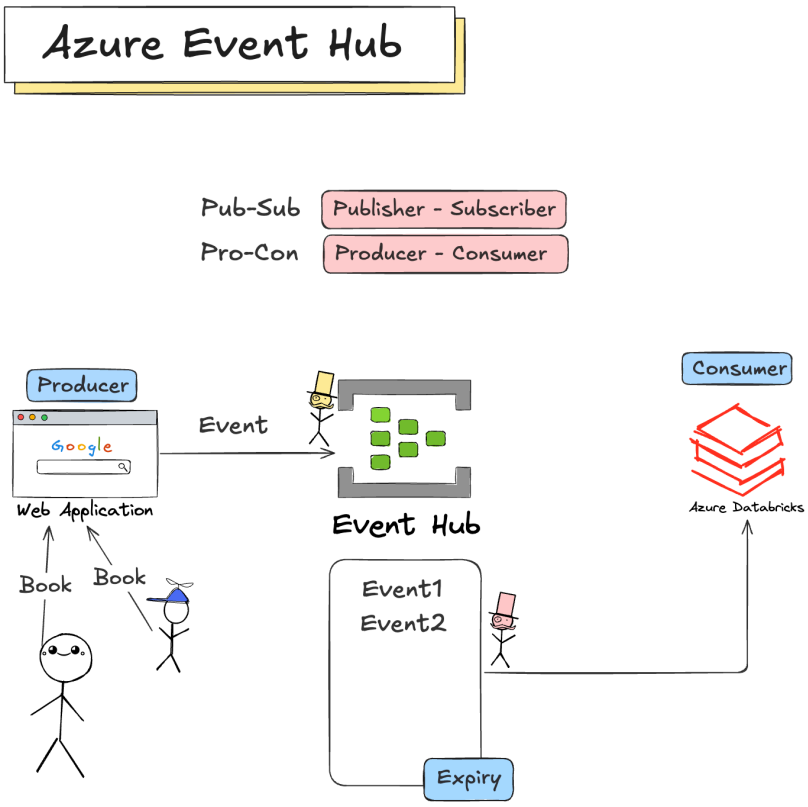
Event Storage and Consumption

Azure Event Hubs provides temporary storage for these streams of events. Once events are stored, various consumers can access and process them.

- **Consumers**: Examples of consumers include:
  - **Azure Databricks**: A powerful analytics platform for processing large datasets.
  - **Azure Synapse Analytics**: An integrated analytics service for data warehousing and Big Data analytics.
  - Other custom applications or services that need to react to real-time events.

Event Retention and Expiry

Events in Azure Event Hubs are not stored indefinitely. They have a configurable **expiry date** or retention period. This means that events are only available for a limited time. Before an event expires, consumers must pull the data and process it. This mechanism ensures that storage is managed efficiently and only relevant, recent data is retained for real-time processing.

Comparison with Apache Kafka

Azure Event Hubs is often described as a cloud-native, fully managed **alternative to Apache Kafka**. While both platforms serve similar purposes as distributed streaming platforms, Event Hubs offers the advantage of being a platform-as-a-service (PaaS) offering from Azure. This means that Microsoft manages the underlying infrastructure, scaling, and maintenance, allowing users to focus solely on their data streaming applications without the operational overhead associated with self-managing Apache Kafka clusters.